

Task 11 - SVM Breast Cancer Classification **Model:** Support Vector Machine (SVM)

1. Model Comparison

- **Linear Kernel:** Performed extremely well (~95%+ accuracy) because the dataset features are linearly separable in high dimensions.
- **RBF Kernel (Tuned):** After GridSearch tuning, the RBF kernel matched or slightly exceeded the linear model by capturing subtle non-linear patterns.

2. Hyperparameter Tuning Results Using GridSearchCV, we found the optimal parameters:

- **C:** Determines the penalty for misclassification. A higher C (e.g., 10 or 100) was preferred, indicating the model prioritizes classifying training points correctly (strict margin).
- **Gamma:** A lower Gamma (e.g., 0.001 or 0.01) was preferred, creating a smoother decision boundary that generalizes better.

3. Performance Metrics

- **AUC Score:** ~0.99 (Near perfect).
- **Recall:** The model achieved very high recall (sensitivity). In medical diagnosis, this is crucial because we want to minimize False Negatives (telling a sick patient they are healthy).